Data Flow Diagrams (DFD) & Data Dictionary

Academic Grievance, No-Dues & Clearance Automation Engine

Standard: Structured Systems Analysis & Design (SSAD) — **Yourdon & DeMarco / Gane & Sarson Notation**
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1. Structured Systems Analysis (SSAD) Notational Guide & Formal Rules

1.1 Notational Conventions & Symbol Guide

This specification adheres strictly to the classical graphical symbols of **Gane & Sarson** and **Yourdon & DeMarco**:

Symbol Element	Gane & Sarson / Yourdon Symbol	Mermaid Semantic Representation	Architectural Definition
External Entity	Double Square / Rectangle	`Entity["External Entity Name"]`	An autonomous origin or destination of data residing outside system boundaries (e.g., Student, Staff).
Process	Rounded Box / Circle	`Process(("ID: Process Name"))`	An algorithmic transformation converting incoming data packets into outgoing information flows.
Data Store	Open-Ended Rectangle	`Store[("DS-ID: Data Store Name")]`	A persistent repository holding system data at rest (e.g., relational database tables).
Data Flow	Directed Vector Arrow	`A -- "Data Flow Name" --> B`	A deterministic pipeline conveying structured data packets between entities, processes, and stores.

1.2 Formal DFD Construction Rules & Quality Compliance

To ensure mathematical and structural consistency, the models conform to the classical **Six Golden Rules of Structured Analysis**:

Structured Analysis Rule	Rule Description & Formal Constraint	Compliance Verification in this Specification
Rule 1: Conservation of Data	A process cannot create data out of nothing ("Miracle") nor consume data without producing output ("Black Hole").	Every process (`1.0` through `6.0`) transforms incoming inputs into concrete, auditable outputs.
Rule 2: Entity-to-Entity Prohibited	External entities cannot communicate directly within the DFD boundary; all interaction must be mediated by a process.	Verified: All student, staff, and admin interactions are processed via numbered transformations.
Rule 3: Store-to-Store Prohibited	Data cannot move directly from one data store to another without executing process manipulation.	Verified: Table interactions (e.g., `DS-4` to `DS-3`) are strictly mediated by processes `2.0` , `3.0` , `4.0` , and `5.0` .
Rule 4: Entity-to-Store Prohibited	External entities cannot directly read or write to data stores without an intervening processing step.	Verified: All database queries and mutations pass through authorized API processes `1.0` to `6.0` .
Rule 5: Horizontal & Vertical Parity	Inputs and outputs of the Level 0 Context diagram must balance perfectly with the aggregate boundary flows at Level 1.	Verified: All 22 boundary flows at Level 0 exhibit 100% exact parity with Level 1 connections.
Rule 6: Decimal Numbering Hierarchy	Unique decimal indexing: Context is `0.0` , Level 1 processes are `1.0` to `6.0` , and Data stores are `DS-1` to `DS-6` .	Verified: Standard hierarchical numbering applied across all sections.

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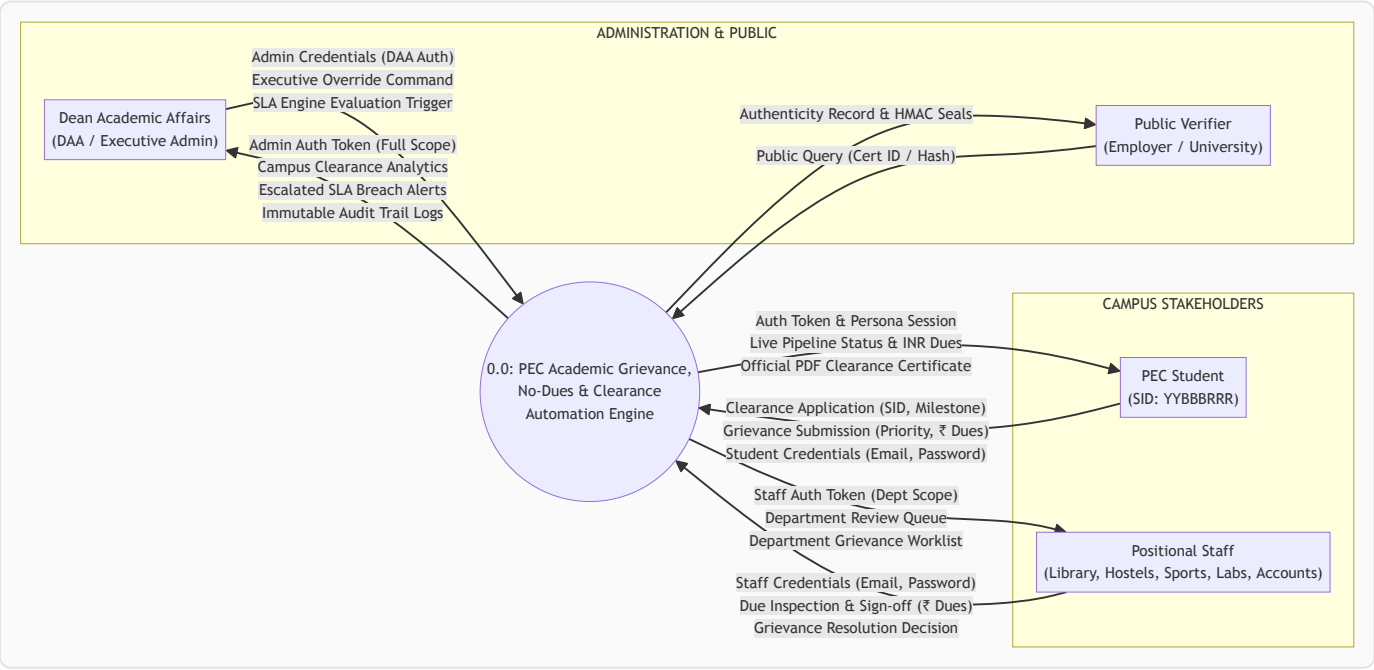
2. DFD Level 0: Context Analysis Diagram

2.1 Environmental Boundary & Entity Characterization

The Context Diagram defines the outermost functional boundary of the **PEC Academic Grievance, No-Dues & Clearance Automation Engine** within **Punjab Engineering College (Sector 12, Chandigarh)**. The central engine (`0.0`) mediates all transactions between four primary external entities:

1. **PEC Student (`Student`):**
   - Identified by an immutable 8-digit PEC SID (`YYBBBRRR` , e.g., `22103042`).
   - Initiates clearance applications, monitors real-time gate progress, views itemized dues in Indian Rupees (`₹`), files grievance appeals, and downloads official tamper-evident PDF certificates.
2. **Positional Department Staff (`Staff`):**
   - Permanent administrative designations: *Librarian* (Dr. B.R. Ambedkar Central Library), *Chief Warden* (PEC Hostels), *President Gymkhana* (Sports Complex), *HOD Labs* (Department Computing & Research Labs), and *Accounts Officer* (Caution Money & Finance Branch).
   - Inspects asset return records, records financial liabilities in INR (`₹`), grants cryptographic sign-offs, and resolves student grievances.
3. **Dean Academic Affairs (`Admin` / DAA):**
   - Chief institutional academic authority overseeing campus-wide clearance compliance.
   - Monitors clearance queues, departmental bottlenecks, and SLA breach escalations; applies executive overrides with mandatory audit justification.
4. **Public Certificate Verifier (`Verifier`):**
   - External employers, background verification agencies, and foreign universities.
   - Validates clearance certificates tokenlessly using Certificate ID or SHA-256 hash digests.

2.2 Level 0 Context Diagram (Mermaid Model)



2.3 Context-Level Input and Output Boundary Data Flows

Flow ID	Flow Name	Direction	Origin	Destination	Data Composition / Packet Description
F-01	`Clearance Application`	Input	`Student`	Process `0.0`	`student_sid` + `academic_year` + `semester` + `reason`
F-02	`Grievance Submission`	Input	`Student`	Process `0.0`	`student_id` + `department_id` + `subject` + `description` + `priority`
F-03	`Student Auth Credentials`	Input	`Student`	Process `0.0`	`email` (`@pec.edu.in`) + `password`
F-04	`Auth Token & Session Scope`	Output	Process `0.0`	`Student`	`access_token` (JWT) + `user_id` + `role: STUDENT` + `roll_number`
F-05	`Live Pipeline Status`	Output	Process `0.0`	`Student`	`overall_status` + 5 $\times$ (`dept_name` , `gate` , `status` , `due_amount` , `remarks`)
F-06	`Grievance Resolution Updates`	Output	Process `0.0`	`Student`	`grievance_id` + `status` + `staff_response` + `resolved_at`
F-07	`Official PDF Certificate`	Output	Process `0.0`	`Student`	Binary vector PDF (`%PDF-`) + DAA/DSA letterhead + SHA-256 seal
F-08	`Staff Auth Credentials`	Input	`Staff`	Process `0.0`	Positional email (`library@pec.edu.in` , etc.) + `password`
F-09	`Due Inspection & Sign-off`	Input	`Staff`	Process `0.0`	`approval_id` + `decision` (`APPROVED` / `REJECTED`) + `remarks` + `due_amount`
F-10	`Grievance Resolution Decision`	Input	`Staff`	Process `0.0`	`grievance_id` + `staff_response` + `status: RESOLVED`
F-11	`Staff Auth Token`	Output	Process `0.0`	`Staff`	`access_token` (JWT) + `role: STAFF` + `department_id`
F-12	`Department Review Queue`	Output	Process `0.0`	`Staff`	List of pending approvals filtered strictly by staff's `department_id`
F-13	`Department Grievance Worklist`	Output	Process `0.0`	`Staff`	List of grievances filtered by `department_id` with active SLA countdowns
F-14	`Admin Auth Credentials`	Input	`Admin`	Process `0.0`	`admin@pec.edu.in` + `password`
F-15	`Executive Override Command`	Input	`Admin`	Process `0.0`	`clearance_id` + `department_code` + `reason` (`ADMIN_OVERRIDE`)
F-16	`SLA Evaluation Trigger`	Input	`Admin`	Process `0.0`	Trigger signal for real-time background SLA deadline evaluation
F-17	`Admin Auth Token`	Output	Process `0.0`	`Admin`	`access_token` (JWT) + `role: ADMIN`
F-18	`Campus Clearance Analytics`	Output	Process `0.0`	`Admin`	Total counts (`approved` , `pending` , `rejected`) + Dept bottleneck hours
F-19	`Escalated Grievance Alerts`	Output	Process `0.0`	`Admin`	List of breached tickets with `is_escalated = True` for DAA action
F-20	`System Audit Trail Logs`	Output	Process `0.0`	`Admin`	Immutable append-only log entries (`actor` , `action` , `target` , `timestamp`)
F-21	`Public Verification Query`	Input	`Verifier`	Process `0.0`	`cert_id` (`PEC-CLR-YYYY-XXXXX`) OR 64-character SHA-256 digest
F-22	`Authenticity Record`	Output	Process `0.0`	`Verifier`	`valid: Boolean` + `student_name` + `sid` + `session` + 5 $\times$ HMAC Seals

3. DFD Level 1: System Decomposition Diagram

3.1 Functional Decomposition Strategy (Processes 1.0 through 6.0)

The Level 1 DFD decomposes the top-level process `0.0` into six dedicated, single-responsibility operational subsystems:

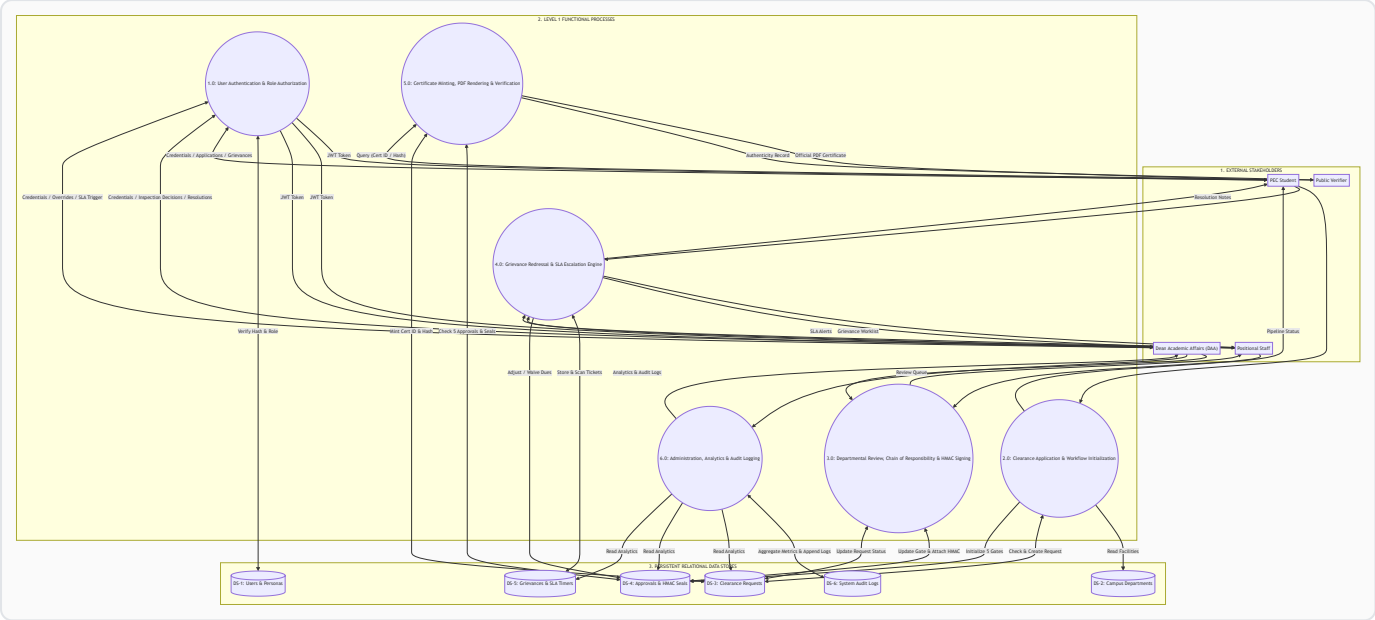
- **`1.0: User Authentication & Role Authorization`**: Validates user credentials, verifies Bcrypt password hashes, issues signed JWT access tokens, and enforces Role-Based Access Control (RBAC).
- **`2.0: Clearance Application & Workflow Initialization`**: Enforces the single-active-request rule, captures student academic milestones, and initializes the 5 departmental approval records across the 3 gates.
- **`3.0: Departmental Review, Chain of Responsibility & HMAC Signing`**: Orchestrates the multi-department clearance pipeline, allows parallel review at Gate 1, enforces prerequisite barriers at Gate 2 and Gate 3, generates HMAC-SHA256 digital seals, and executes DAA overrides.
- **`4.0: Grievance Redressal & SLA Escalation Engine`**: Manages student disputes, tags priority SLA turnaround windows (4h, 12h, 24h, 48h), evaluates deadline breaches, and auto-escalates delinquent tickets to the DAA.
- **`5.0: Certificate Minting, ReportLab PDF Rendering & Public Verification`**: Mints unique certificate identifiers upon Gate 3 completion, releases the ₹10,000.00 Caution Money deposit, computes SHA-256 verification digests, renders vector PDF certificates, and serves tokenless public authenticity verifications.
- **`6.0: Institutional Administration, Analytics & Audit Logging`**: Computes facility bottleneck statistics, provides institutional oversight for the DAA, and immutably persists all lifecycle transactions into an append-only audit trail.

3.2 Persistent Data Stores Overview (`DS-1` through `DS-6`)

The system state is maintained across six relational data stores:

- **`DS-1: Users & Personas`** (`users` table): User identities, authentication hashes, roles (`STUDENT`, `STAFF`, `ADMIN`), and department linkages.
- **`DS-2: Campus Departments`** (`departments` table): Academic facilities, institutional codes, gate assignments, and staff head designations.
- **`DS-3: Clearance Requests`** (`clearance_requests` table): Master student clearance records, milestone metadata, overall statuses, and certificate digests.
- **`DS-4: Department Approvals & HMAC Seals`** (`department_approvals` table): Departmental review records, itemized INR dues, reviewer remarks, and HMAC-SHA256 digital seals.
- **`DS-5: Grievances & SLA Timers`** (`grievances` table): Disputed dues tickets, priority classes, computed SLA deadlines, and escalation flags.
- **`DS-6: System Audit Logs`** (`audit_logs` table): Append-only audit events tracking actors, actions, timestamps, and JSON-encoded metadata.

3.3 Level 1 System Decomposition Diagram (Mermaid Model)



3.4 Level 1 Inter-Process, Store, and Entity Data Flows

Process ID	Process Name	Consumed Inputs (From Entities & Stores)	Generated Outputs (To Entities & Stores)
`1.0`	**Authentication & Role Authorization**	`Student Credentials`, `Staff Credentials`, `Admin Credentials`, `DS-1: users` password hashes.	`JWT Access Tokens` (Student, Staff, Admin scopes with claims: `sub`, `role`, `exp`).
`2.0`	**Clearance Application & Initialization**	`Clearance Application` from Student, `DS-3: clearance_requests` (duplicate check), `DS-2: departments`.	Created record in `DS-3`, 5 initialized approval rows in `DS-4`, audit event to `6.0`, `Pipeline Status` to Student.
`3.0`	**Review, Chain of Responsibility & HMAC Signing**	`Due Inspection & Sign-off` from Staff, `Executive Override` from Admin, current approval states from `DS-4`.	Updated status & `HMAC Seal` in `DS-4`, unlocked downstream gates in `DS-4`, request status in `DS-3`, audit event to `6.0`.
`4.0`	**Grievance Redressal & SLA Escalation**	`Grievance Submission` from Student, `Resolution Decision` from Staff, `SLA Trigger` from Admin, deadline scans on `DS-5`.	Ticket recorded in `DS-5`, updated dues in `DS-4`, `SLA Breach Alerts` to Admin, resolution notes to Student, audit event to `6.0`.
`5.0`	**Certificate Minting & Verification**	Verified approvals from `DS-4`, cleared request from `DS-3`, `Verification Query` from Verifier.	`Cert ID` and `SHA-256 Digest` stored in `DS-3`, `Official PDF` streamed to Student, `Authenticity Record` to Verifier.
`6.0`	**Admin Analytics & Audit Logging**	Internal audit events from `2.0`, `3.0`, `4.0`, `5.0`; analytical aggregations across `DS-3`, `DS-4`, `DS-5`, `DS-6`.	Append-only records to `DS-6`, `Campus Analytics` & `Bottleneck Metrics` to Admin, `Audit Trail Logs` to Admin.

4. Process Specifications (P-Specs), Decision Tables & Decision Trees

4.1 Process 3.0: Chain of Responsibility (CoR) Gating & Multi-Signatory Barrier Logic

Process Narrative & Algorithmic Specification:

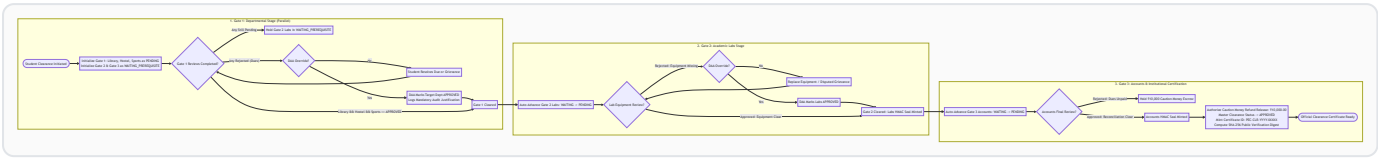
Process `3.0` coordinates the multi-department approval workflow using the behavioral **Chain of Responsibility** pattern (`backend/app/patterns/chain_of_responsibility.py`). It enforces strict barrier rules:

1. **Gate 1 (Parallel Gates):** Dr. B.R. Ambedkar Central Library, PEC Hostels, and Gymkhana evaluate clearances concurrently. Neither blocks the other.
2. **Gate 2 (Sequential Dependency Barrier):** Departmental Laboratories (CSE/ECE/Mech/etc.) verify hardware and lab equipment returns **strictly after all Gate 1 facilities are APPROVED**. Any review attempt before Gate 1 is 100% complete is rejected with `Prerequisite Gate Error`.
3. **Gate 3 (Final Gate Barrier):** Accounts Branch reconciles caution money and fee balances **strictly after Gates 1 and 2 are 100% APPROVED**.
4. **Cryptographic Sealing:** An approved review generates an irreversible HMAC-SHA256 signature seal: $$\text{Digital Signature} = \text{HMAC-SHA256}(K_{\text{secret}}, \text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp})$$
5. **Caution Money Disbursement:** Approval at Gate 3 unblocks the release authorization of the student's institutional Caution Money deposit (₹10,000.00).

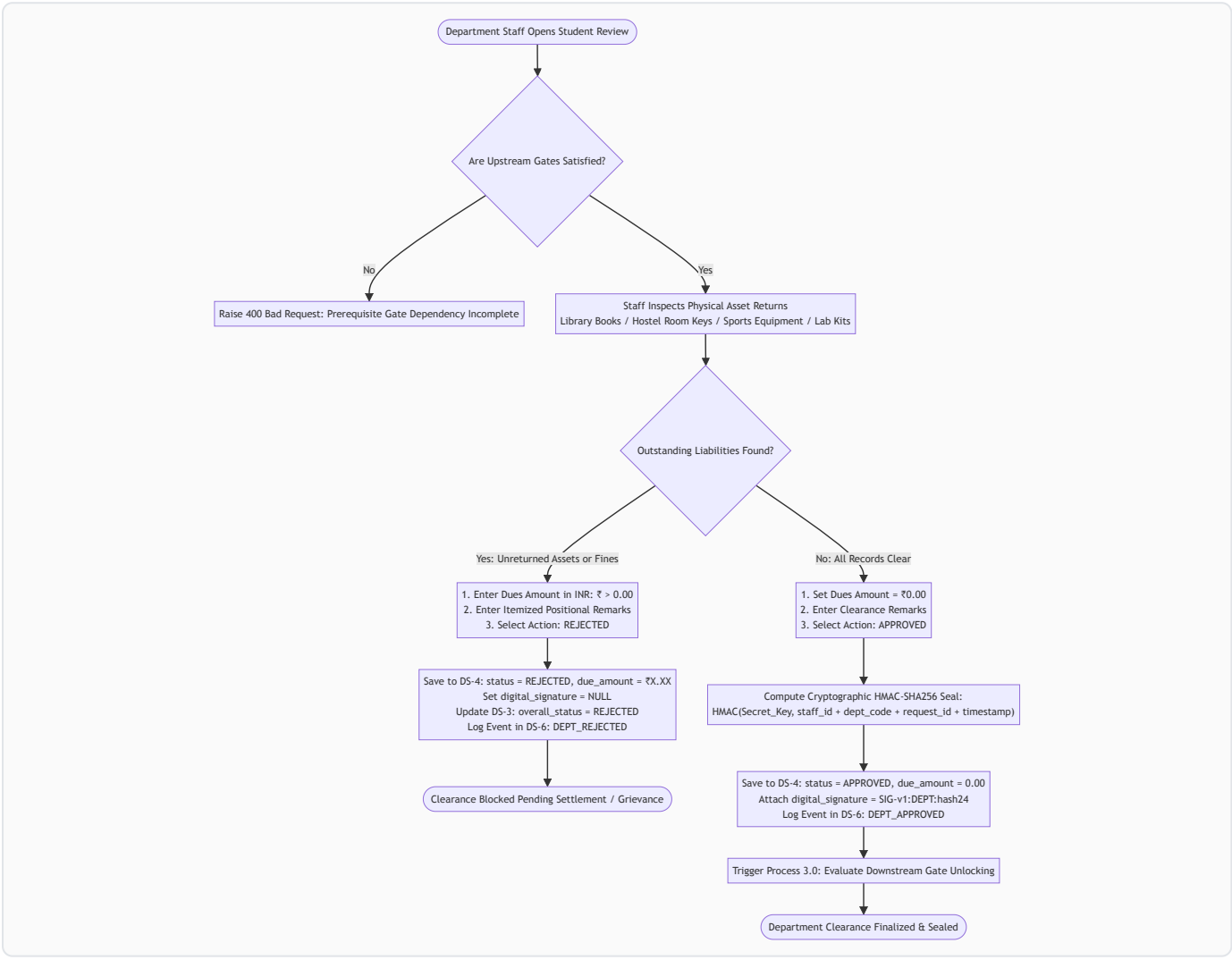
Decision Table 1: Multi-Gate Clearance Pipeline & Prerequisite Barrier Matrix

Rule ID	Gate 1 Status (Lib / Host / Sport)	Gate 2 Status (Dept Labs)	Gate 3 Status (Accounts)	DAA Override?	Action Target & Enforced State Transition
R-101	All `PENDING`	`WAITING`	`WAITING`	No	Allow concurrent Gate 1 review; update to `APPROVED` or `REJECTED`; Mint HMAC seal on approval.
R-102	Any `PENDING` / `REJECTED`	`WAITING`	`WAITING`	No	**BLOCK GATE 2 WITH 400 ERROR:** Prerequisite Gate 1 facilities incomplete or disputed.
R-103	Any `REJECTED` (Dues > ₹0)	`WAITING`	`WAITING`	No	**REJECT REQUEST:** Master status `REJECTED`; student must settle dues or file grievance.
R-104	All `APPROVED`	`WAITING`	`WAITING`	No	**AUTO-UNLOCK GATE 2:** Automatically advance Gate 2 (Dept Labs) from `WAITING` to `PENDING`.
R-105	All `APPROVED`	`PENDING`	`WAITING`	No	Allow HOD Labs review; if approved, mint HMAC seal and update status to `APPROVED`.
R-106	All `APPROVED`	`APPROVED`	`WAITING`	No	**AUTO-UNLOCK GATE 3:** Automatically advance Gate 3 (Accounts) from `WAITING` to `PENDING`.
R-107	All `APPROVED`	`APPROVED`	`PENDING`	No	Allow Accounts review; if approved, authorize release of ₹10,000 Caution Money deposit refund.
R-108	All `APPROVED`	`APPROVED`	`APPROVED`	No	**MASTER CLEARANCE GRANTED:** Master status `APPROVED`; Mint Certificate ID & SHA-256 Digest.
R-109	Any `REJECTED`	Any	Any	**Yes (DAA)**	**EXECUTIVE OVERRIDE:** Mark target department `APPROVED`; clear dues; log `ADMIN_OVERRIDE` in `DS-6`.

Decision Tree 1: Clearance Gating & Barrier Enforcement Tree (Mermaid)



Decision Tree 2: Departmental Liability Assessment & HMAC Signing Flow (Mermaid)



4.2 Process 4.0: Grievance Redressal & SLA Escalation Engine

Process Narrative & Algorithmic Specification:

Process `4.0` governs student appeals regarding disputed fines (e.g., contested hostel mess bills or library replacement costs). It enforces dynamic Service Level Agreements (SLA):

1. **Dynamic Deadline Calculation:** $$\text{SLA Deadline Timestamp} = \text{Submission Timestamp} + \text{SLA Window}(\text{Priority})$$

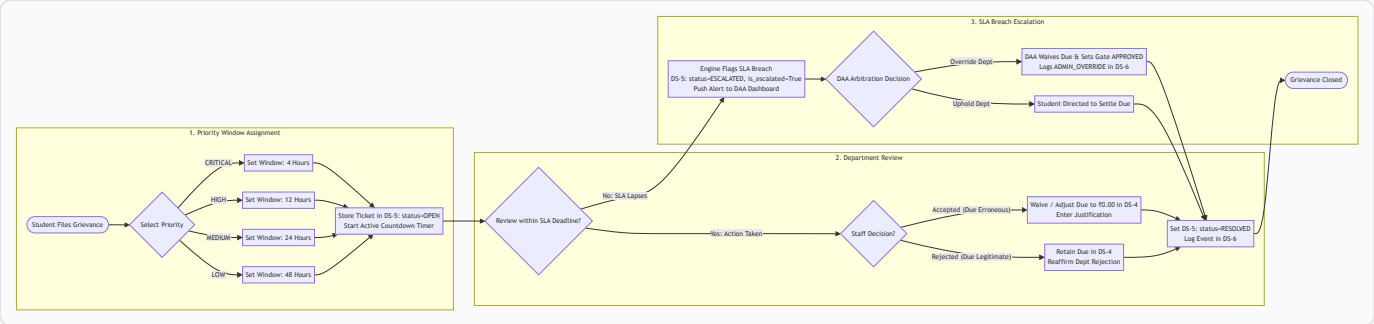
- CRITICAL Priority: **4 Hours**
- HIGH Priority: **12 Hours**
- MEDIUM Priority: **24 Hours**
- LOW Priority: **48 Hours**

2. **Real-time Escalation Scanner:** Evaluates all tickets where `Current_Time > sla_deadline` and `status in ('OPEN', 'IN_REVIEW')`. Expired tickets are updated to `status = 'ESCALATED'` with `is_escalated = True`.
3. **Executive Alerts:** Breached tickets generate immediate visual escalation banners on the Dean Academic Affairs dashboard.
4. **Dispute Settlement:** Resolution by staff or DAA updates the ticket to `RESOLVED`, sets `resolved_at`, adjusts or waives the disputed fine in `DS-4`, and logs the settlement in `DS-6`.

Decision Table 2: Priority SLA Deadline & Automated Escalation Matrix

Rule ID	Priority	Allowed SLA	Elapsed Time (t)	Reviewer Action	Enforced System Action & Ticket Mutation
R-201	`CRITICAL`	**4 Hours**	$t \le$ 4 Hours	Pending review	Maintain live countdown timer (`status = OPEN`, `is_escalated = False`).
R-202	`CRITICAL`	**4 Hours**	**$t >$ 4 Hours**	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach; set `status = ESCALATED`, `is_escalated = True`.
R-203	`HIGH`	**12 Hours**	$t \le$ 12 Hours	Pending review	Maintain live countdown timer (`status = OPEN`, `is_escalated = False`).
R-204	`HIGH`	**12 Hours**	**$t >$ 12 Hours**	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach; set `status = ESCALATED`, `is_escalated = True`.
R-205	`MEDIUM`	**24 Hours**	$t \le$ 24 Hours	Pending review	Maintain live countdown timer (`status = OPEN`, `is_escalated = False`).
R-206	`MEDIUM`	**24 Hours**	**$t >$ 24 Hours**	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach; set `status = ESCALATED`, `is_escalated = True`.
R-207	`LOW`	**48 Hours**	$t \le$ 48 Hours	Pending review	Maintain live countdown timer (`status = OPEN`, `is_escalated = False`).
R-208	`LOW`	**48 Hours**	**$t >$ 48 Hours**	None taken	**TRIGGER AUTO-ESCALATION:** Flag ticket for DAA; log breach; set `status = ESCALATED`, `is_escalated = True`.
R-209	Any	Any	Any	Staff/DAA registers resolution	Set `status = RESOLVED`, record `staff_response`, zero or adjust INR dues in `DS-4`, set `resolved_at = Now`.

Decision Tree 3: Grievance Arbitration & SLA Breach Escalation Tree (Mermaid)



4.3 Process 1.0 & 3.0: Role-Based Access Control (RBAC) & Authority Validation

Process Narrative & Algorithmic Specification:

The API Gateway enforces strict **Role-Based Access Control (RBAC)** across four user scopes: `STUDENT`, `STAFF`, `ADMIN` (DAA), and unauthenticated `PUBLIC`. Unauthorized requests are blocked with HTTP `401 Unauthorized` or `403 Forbidden`.

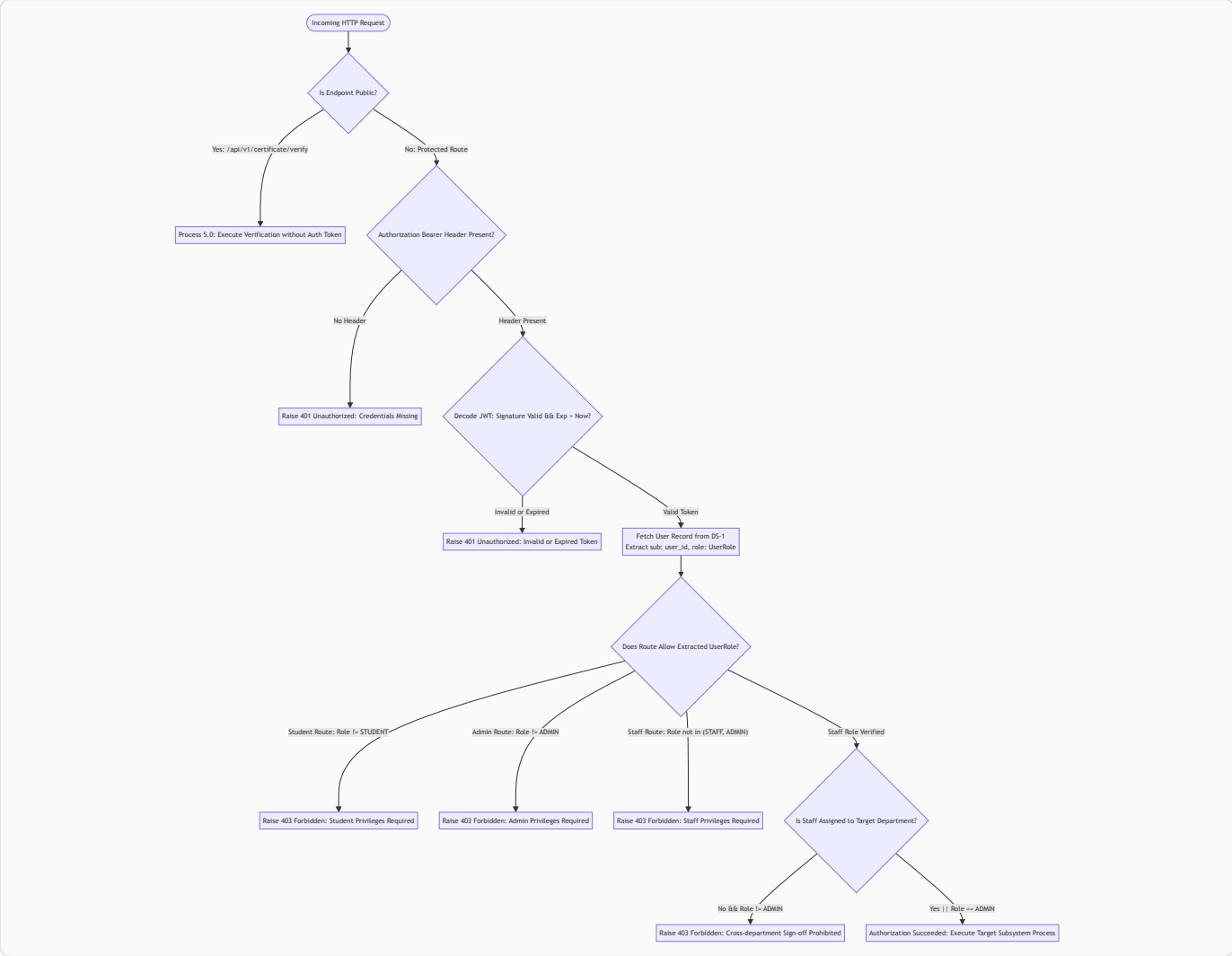
Decision Table 3: Comprehensive RBAC Authorization Matrix

Action / Operation	API Route Endpoint	`STUDENT`	`STAFF` (Own)	`STAFF` (Other)	`ADMIN`	`PUBLIC`	Enforced Breach Error
Apply for Clearance	`POST /api/v1/clearance/apply`	**ALLOW**	DENY	DENY	DENY	DENY	`403 Forbidden`
View My Pipeline	`GET /api/v1/clearance/my-requests`	**ALLOW**	DENY	DENY	DENY	DENY	`401 / 403`
View Staff Queue	`GET /api/v1/clearance/staff/queue`	DENY	**ALLOW**	DENY	**ALLOW**	DENY	`403 Forbidden`
Sign Department Gate	`POST /api/v1/clearance/staff/decision/{id}`	DENY	**ALLOW**	DENY	**ALLOW**	DENY	`403 Forbidden`
File Grievance Ticket	`POST /api/v1/grievance`	**ALLOW**	DENY	DENY	DENY	DENY	`403 Forbidden`
Resolve Grievance	`POST /api/v1/grievance/{id}/resolve`	DENY	**ALLOW**	DENY	**ALLOW**	DENY	`403 Forbidden`
Trigger SLA Scan	`POST /api/v1/grievance/trigger-sla-check`	DENY	**ALLOW**	**ALLOW**	**ALLOW**	DENY	`403 Forbidden`
View Campus Analytics	`GET /api/v1/admin/analytics`	DENY	**ALLOW**	**ALLOW**	**ALLOW**	DENY	`403 Forbidden`
View System Audit Logs	`GET /api/v1/admin/audit-logs`	DENY	DENY	DENY	**ALLOW**	DENY	`403 Forbidden`
Execute Admin Override	`POST /api/v1/admin/clearance/{id}/override`	DENY	DENY	DENY	**ALLOW**	DENY	`403 Forbidden`
Download PDF Cert	`GET /api/v1/certificate/download/{id}`	**ALLOW**	**ALLOW**	**ALLOW**	**ALLOW**	DENY	`401 Unauthorized`

Action / Operation	API Route Endpoint	`STUDENT`	`STAFF` (Own)	`STAFF` (Other)	`ADMIN`	`PUBLIC`	Enforced Breach Error
Public Verification	`GET /api/v1/certificate/verify`	**ALLOW**	**ALLOW**	**ALLOW**	**ALLOW**	**ALLOW**	Public Access

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Decision Tree 4: User Authentication & Authorization Gate Tree (Mermaid)



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5. Comprehensive Data Dictionary

5.1 Data Stores at Rest (`DS-1` to `DS-6` Schemas)

The schemas reflect the exact production relational database definitions (`backend/app/models/models.py`):

`DS-1`: Users & Personas (`users` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Auto-incrementing surrogate identifier	`1`

Column Name	SQL Type	Constraint	Description	Concrete Example
`email`	`VARCHAR(120)`	Unique, Not Null	Institutional PEC email address	`alice@pec.edu.in`
`full_name`	`VARCHAR(120)`	Not Null	User full name or administrative title	`Alice Sharma` / `Chief Warden`
`role`	`ENUM`	Not Null	Security role: `STUDENT`, `STAFF`, `ADMIN`	`STUDENT`
`roll_number`	`VARCHAR(50)`	Unique, Nullable	8-digit PEC SID format (`YYBBBRRR`)	`22103042`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	Foreign key linking staff to their facility	`1` (Library)
`hashed_password`	`VARCHAR(255)`	Not Null	Salted Bcrypt hash (work factor 12)	`$2b$12$e8x...`
`created_at`	`DATETIME`	Default UTC	Timestamp of user provisioning	`2026-05-01 09:00:00`

`DS-2` : Campus Departments (`departments` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Department surrogate identifier	`1`
`name`	`VARCHAR(100)`	Not Null	Official campus facility designation	`Dr. B. R. Ambedkar Central Library`
`code`	`VARCHAR(50)`	Unique, Not Null	Institutional department code	`LIBRARY`, `HOSTEL`, `SPORTS`, `LABS`, `ACCOUNTS`
`clearance_gate`	`INTEGER`	Not Null	Pipeline gate level: `1` (Parallel), `2` (Labs), `3` (Accounts)	`1`
`description`	`VARCHAR(255)`	Nullable	Detailed functional remit	`Central university library textbook dues`
`staff_head_name`	`VARCHAR(100)`	Nullable	Positional title of head signatory	`Chief Librarian`

`DS-3` : Clearance Requests (`clearance_requests` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Clearance request unique identifier	`42`
`student_id`	`INTEGER`	Foreign Key (`users.id`)	References applicant student	`1` (Alice Sharma)
`academic_year`	`VARCHAR(20)`	Not Null	Enrolled academic year session	`2025-2026`
`semester`	`VARCHAR(20)`	Not Null	Current academic semester	`Semester 8`
`reason`	`VARCHAR(100)`	Not Null	Institutional justification	`Graduation Clearance & Degree Award`
`overall_status`	`ENUM`	Not Null	Master status: `PENDING`, `IN_PROGRESS`, `APPROVED`, `REJECTED`	`APPROVED`
`certificate_id`	`VARCHAR(64)`	Unique, Nullable	Minted certificate number	`PEC-CLR-2026-00042`
`certificate_hash`	`VARCHAR(128)`	Nullable	SHA-256 public verification digest	`7f83b1657ff1fc53b92dc18148a1d65dfc...`
`created_at`	`DATETIME`	Default UTC	Timestamp of clearance application	`2026-05-01 10:00:00`
`updated_at`	`DATETIME`	On Update UTC	Timestamp of last state mutation	`2026-05-03 14:30:00`

DS-4 : Department Approvals (`department_approvals` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Approval record surrogate key	`201`
`request_id`	`INTEGER`	Foreign Key (`clearance_requests.id`)	References parent clearance request	`42`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	References reviewing department	`1` (Library)
`status`	`ENUM`	Not Null	`WAITING_PREREQUISITE` , `PENDING` , `APPROVED` , `REJECTED`	`APPROVED`
`staff_id`	`INTEGER`	Foreign Key (`users.id`)	References signing staff official	`2` (Librarian)
`approved_at`	`DATETIME`	Nullable	Exact timestamp of review sign-off	`2026-05-02 11:15:00`
`remarks`	`TEXT`	Nullable	Itemized inspection commentary	`All library books returned. Zero dues.`
`due_amount`	`FLOAT`	Default 0.0	Outstanding liability in INR (`₹`)	`0.00`
`digital_signature`	`VARCHAR(255)`	Nullable	Keyed HMAC-SHA256 signature seal	`SIG-v1:LIBRARY:8a7f9c2d1b4e5f...`

DS-5 : Grievances & SLA Timers (`grievances` table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Grievance ticket identifier	`8`
`student_id`	`INTEGER`	Foreign Key (`users.id`)	References appellant student	`3` (Charlie Verma)
`clearance_request_id`	`INTEGER`	Foreign Key (`clearance_requests.id`)	Associated clearance application	`42`
`department_id`	`INTEGER`	Foreign Key (`departments.id`)	Target disputed department	`2` (Hostels)
`subject`	`VARCHAR(200)`	Not Null	Summary dispute title	`Shivalik Mess Dues Inaccuracy`
`description`	`TEXT`	Not Null	Detailed appeal justification	`Internship sanctioned by DAA for Jan-June`
`priority`	`ENUM`	Not Null	`LOW` , `MEDIUM` , `HIGH` , `CRITICAL`	`HIGH`
`status`	`ENUM`	Not Null	`OPEN` , `IN_REVIEW` , `RESOLVED` , `ESCALATED`	`ESCALATED`
`sla_hours`	`INTEGER`	Default 24	Allowed resolution window in hours	`12`
`sla_deadline`	`DATETIME`	Not Null	Computed absolute expiration deadline	`2026-05-02 22:00:00`
`is_escalated`	`BOOLEAN`	Default False	True if deadline passed without resolution	`True`

Column Name	SQL Type	Constraint	Description	Concrete Example
`staff_response`	`TEXT`	Nullable	Final written response from authority	`Mess rebate verified; dues waived.`
`created_at`	`DATETIME`	Default UTC	Submission timestamp	`2026-05-02 10:00:00`
`resolved_at`	`DATETIME`	Nullable	Exact timestamp of resolution	`2026-05-03 08:30:00`

`DS-6` : System Audit Logs (**`audit_logs`** table)

Column Name	SQL Type	Constraint	Description	Concrete Example
`id`	`INTEGER`	Primary Key	Sequential audit entry index	`1052`
`actor_id`	`INTEGER`	Nullable	User ID of the mutating actor	`2`
`actor_name`	`VARCHAR(120)`	Not Null	Human-readable name of the actor	`Chief Librarian` / `SLA_ENGINE`
`actor_role`	`VARCHAR(50)`	Not Null	Security role: `STUDENT` , `STAFF` , `ADMIN` , `SYSTEM`	`STAFF`
`action`	`VARCHAR(100)`	Not Null	Immutable action verb	`LIBRARY_APPROVED` , `ADMIN_OVERRIDE`
`target_type`	`VARCHAR(50)`	Not Null	Entity category: `CLEARANCE` , `GRIEVANCE`	`CLEARANCE`
`target_id`	`INTEGER`	Nullable	Foreign ID of the affected entity	`42`
`details`	`TEXT`	Nullable	Textual metadata and mutation payload	`Decision: APPROVED. Dues: ₹0.00. Sig: SIG-v1...`
`timestamp`	`DATETIME`	Default UTC	Microsecond precision timestamp	`2026-05-02 11:15:02`

5.2 Data Elements in Motion (Composite Packets & Primitive Elements)

Composite Data Packets:

1. **`Clearance_Submission_Packet`** : $$\text{Packet} = {\text{student_sid: String}, \text{academic_year: String}, \text{semester: String}, \text{reason: String}}$$
2. **`Department_Signoff_Packet`** : $$\text{Packet} = {\text{approval_id: Int}, \text{decision: Enum}, \text{remarks: String}, \text{due_amount: Float} \ge 0.00}$$
3. **`Grievance_Filing_Packet`** : $$\text{Packet} = {\text{department_id: Int}, \text{subject: String}, \text{description: String}, \text{priority: Enum}}$$
4. **`Public_Verification_Result_Packet`** : $$\text{Packet} = {\text{valid: Bool}, \text{cert_id: String}, \text{student_name: String}, \text{sid: String}, \text{signatures: Array[HMAC_Seal]}}$$

Primitive Data Elements:

- **`SID`** **(Student Identification Number):**
  - **Type:** String (8 digits).
  - **Regular Expression:** `^[0-9]{8}$` (e.g., `22103042`).
  - **Domain:** Valid Punjab Engineering College student enrollment codes.

- `due_amount` **(Departmental Liability):**
  - **Type:** Float (2 decimal precision).
  - **Currency:** Indian Rupee (`₹` / INR).
  - **Domain:** `[0.00, 500000.00]`. Negative numbers are rejected by schema validators with `422 Unprocessable Entity`.
- `digital_signature` **(HMAC Approval Seal):**
  - **Type:** String (ASCII, formatted).
  - **Syntax:** `SIG-v1:<DEPT_CODE>:<24_CHAR_HEX>` (e.g., `SIG-v1:LIBRARY:a7f9b2c3d4e5f6018293a4b5`).
  - **Security Property:** Non-repudiable; invalidates if staff ID, department, request ID, or timestamp are altered.
- `certificate_hash` **(Public Verification Digest):**
  - **Type:** String (64-character lowercase hexadecimal).
  - **Formula:** `SHA-256(cert_id + ":" + student_id + ":" + academic_year + ":" + semester)`

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6. Formal Horizontal & Vertical Balancing Verification

To ensure that the DFD specifications contain zero structural discrepancies and are frozen permanently, the table below verifies horizontal and vertical parity between Level 0 and Level 1:

Level 0 Boundary Flow Index	Boundary Flow Name	External Entity	Level 0 Direction	Level 1 Connecting Subsystem Process	Balancing Parity Status
F-01	`Clearance Application`	`Student`	Input	Process `2.0: Clearance Application & Workflow`	**BALANCED (100%)**
F-02	`Grievance Submission`	`Student`	Input	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**
F-03	`Student Auth Credentials`	`Student`	Input	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-04	`Auth Token & Session Scope`	`Student`	Output	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-05	`Live Pipeline Status`	`Student`	Output	Process `2.0: Clearance Application & Workflow`	**BALANCED (100%)**
F-06	`Grievance Resolution Updates`	`Student`	Output	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**
F-07	`Official PDF Certificate`	`Student`	Output	Process `5.0: Certificate Minting & Verification`	**BALANCED (100%)**
F-08	`Staff Auth Credentials`	`Staff`	Input	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-09	`Due Inspection & Sign-off`	`Staff`	Input	Process `3.0: Review, Chain of Responsibility & HMAC`	**BALANCED (100%)**
F-10	`Grievance Resolution Decision`	`Staff`	Input	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**
F-11	`Staff Auth Token`	`Staff`	Output	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-12	`Department Review Queue`	`Staff`	Output	Process `3.0: Review, Chain of Responsibility & HMAC`	**BALANCED (100%)**
F-13	`Department Grievance Worklist`	`Staff`	Output	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**

Level 0 Boundary Flow Index	Boundary Flow Name	External Entity	Level 0 Direction	Level 1 Connecting Subsystem Process	Balancing Parity Status
F-14	`Admin Auth Credentials`	`Admin`	Input	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-15	`Executive Override Command`	`Admin`	Input	Process `3.0: Review, Chain of Responsibility & HMAC`	**BALANCED (100%)**
F-16	`SLA Evaluation Trigger`	`Admin`	Input	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**
F-17	`Admin Auth Token`	`Admin`	Output	Process `1.0: Authentication & Role Authorization`	**BALANCED (100%)**
F-18	`Campus Clearance Analytics`	`Admin`	Output	Process `6.0: Administration, Analytics & Logging`	**BALANCED (100%)**
F-19	`Escalated Grievance Alerts`	`Admin`	Output	Process `4.0: Grievance Redressal & SLA Engine`	**BALANCED (100%)**
F-20	`System Audit Trail Logs`	`Admin`	Output	Process `6.0: Administration, Analytics & Logging`	**BALANCED (100%)**
F-21	`Public Verification Query`	`Verifier`	Input	Process `5.0: Certificate Minting & Verification`	**BALANCED (100%)**
F-22	`Authenticity Record & Seals`	`Verifier`	Output	Process `5.0: Certificate Minting & Verification`	**BALANCED (100%)**

Conclusion: All 22 boundary data flows exhibit exact horizontal and vertical parity between Level 0 and Level 1. The data stores (`DS–1` through `DS–6`) match production relational tables, decision tables cover all operational branching conditions, decision trees visually articulate complex state lifecycles, and no lower-level decompositions exceed Level 1. This document constitutes the **Frozen Final Baseline** of the Structured Systems Analysis for the Punjab Engineering College Clearance Automation Engine.